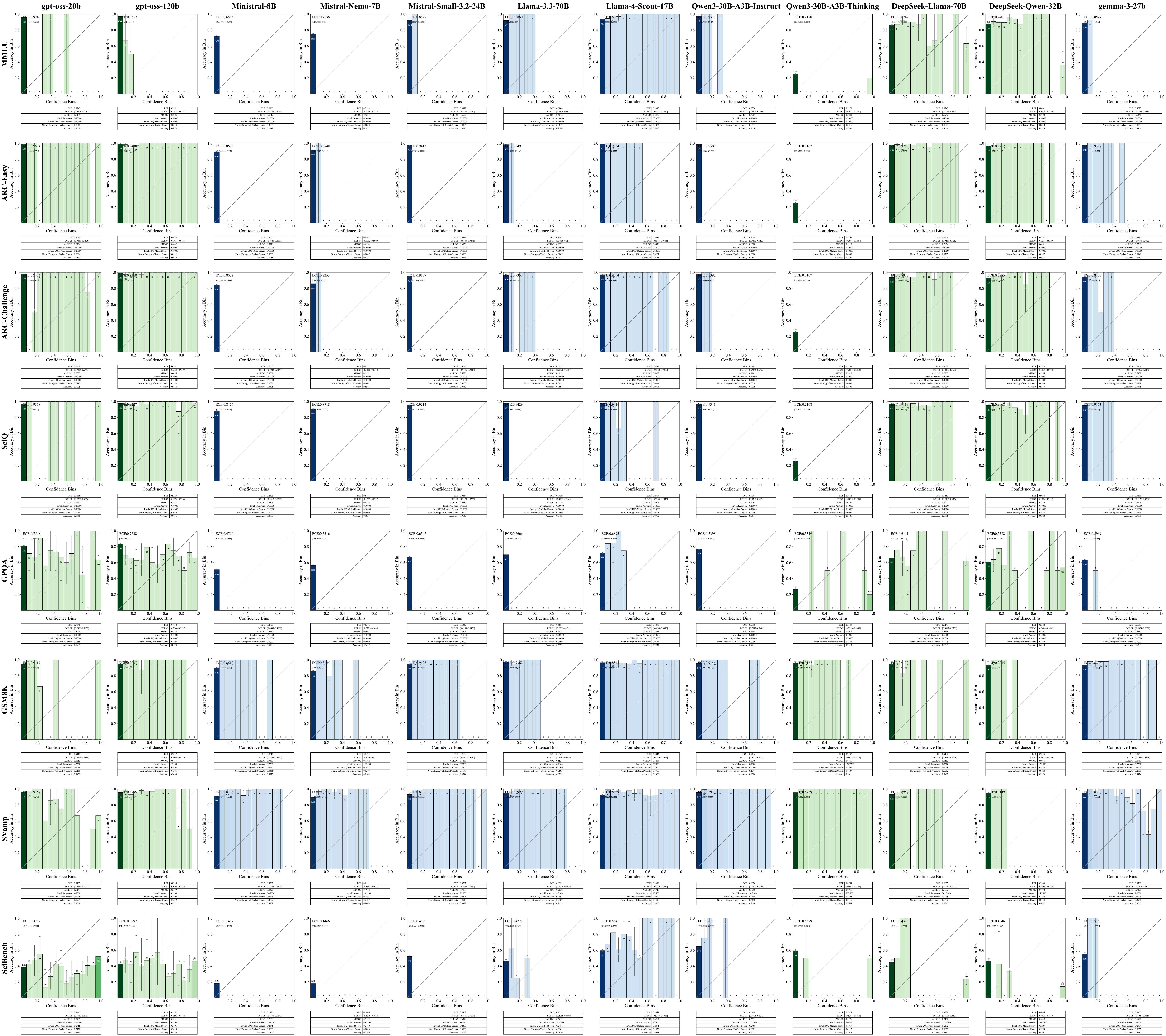
Calibration Plots for Uncertainty Metric CCP



MMLU

	gpt-oss-20b	gpt-oss-120b	Ministral-8B	Mistral-Nemo-7B	Mistral-Small-3.2-24B	Llama-3.3-70B	Llama-4-Scout-17B	Qwen3-30B-A3B-Instruct	Qwen3-30B-A3B-Thinking	DeepSeek-Llama-70B	DeepSeek-Qwen-32B	gemma-3-27b
ECE	0.9243	0.9352	0.6885	0.7138	0.8877	0.8860	0.8933	0.9376	0.2170	0.8242	0.8401	0.8527
ECE CI	[0.9203–0.9282]	[0.9318–0.9381]	[0.6796–0.6964]	[0.7050–0.7226]	[0.8829–0.8932]	[0.8806–0.8911]	[0.8883–0.8980]	[0.9343–0.9409]	[0.2087–0.2250]	[0.8176–0.8308]	[0.8336–0.8464]	[0.8466–0.8589]
AUROC	0.6135	0.5005	0.5812	0.5633	0.6652	0.6846	0.6385	0.6287	0.6230	0.5942	0.5708	0.6440
Invalid Answers	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000
Invalid UQ Method Scores	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000
Norm. Entropy of Bucket Counts	0.0011	0.0043	0.0000	0.0000	0.0004	0.0057	0.1083	0.0011	0.0016	0.1001	0.0810	0.0039
Accuracy	0.9578	0.9684	0.7218	0.7471	0.9210	0.9196	0.9360	0.9710	0.2500	0.8648	0.8776	0.8861

ARC-Easy

	gpt-oss-20b	gpt-oss-120b	Ministral-8B	Mistral-Nemo-7B	Mistral-Small-3.2-24B	Llama-3.3-70B	Llama-4-Scout-17B	Qwen3-30B-A3B-Instruct	Qwen3-30B-A3B-Thinking	DeepSeek-Llama-70B	DeepSeek-Qwen-32B	gemma-3-27b
ECE	0.9514	0.9438	0.8605	0.8848	0.9413	0.9491	0.9534	0.9509	0.2167	0.9250	0.9252	0.9392
ECE CI	[0.9488–0.9536]	[0.9414–0.9465]	[0.8548–0.8667]	[0.8792–0.8900]	[0.9383–0.9441]	[0.9466–0.9516]	[0.9513–0.9555]	[0.9483–0.9533]	[0.2084–0.2249]	[0.9218–0.9283]	[0.9214–0.9287]	[0.9358–0.9422]
AUROC	0.6734	0.4442	0.5779	0.6114	0.6820	0.6323	0.6830	0.8308	0.5910	0.5854	0.6081	0.7588
Invalid Answers	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000
Invalid UQ Method Scores	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000
Norm. Entropy of Bucket Counts	0.0096	0.0912	0.0000	0.0007	0.0000	0.0007	0.0227	0.0000	0.0000	0.1797	0.0957	0.0148
Accuracy	0.9862	0.9936	0.8938	0.9181	0.9746	0.9824	0.9878	0.9842	0.2500	0.9748	0.9653	0.9732

ARC-Challenge

	gpt-oss-20b	gpt-oss-120b	Ministral-8B	Mistral-Nemo-7B	Mistral-Small-3.2-24B	Llama-3.3-70B	Llama-4-Scout-17B	Qwen3-30B-A3B-Instruct	Qwen3-30B-A3B-Thinking	DeepSeek-Llama-70B	DeepSeek-Qwen-32B	gemma-3-27b
ECE	0.9424	0.9368	0.8072	0.8253	0.9177	0.9357	0.9354	0.9395	0.2167	0.8928	0.8885	0.9106
ECE CI	[0.9394–0.9453]	[0.9338–0.9397]	[0.8003–0.8144]	[0.8186–0.8318]	[0.9136–0.9215]	[0.9320–0.9387]	[0.9320–0.9385]	[0.9366–0.9425]	[0.2085–0.2253]	[0.8880–0.8976]	[0.8830–0.8938]	[0.9059–0.9150]
AUROC	0.6333	0.6435	0.5659	0.5512	0.6690	0.6058	0.6303	0.7141	0.5827	0.5977	0.5756	0.6459
Invalid Answers	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000
Invalid UQ Method Scores	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000
Norm. Entropy of Bucket Counts	0.0110	0.1124	0.0000	0.0007	0.0000	0.0021	0.0357	0.0014	0.0000	0.1512	0.0845	0.0139
Accuracy	0.9775	0.9878	0.8405	0.8586	0.9510	0.9691	0.9713	0.9729	0.2500	0.9394	0.9277	0.9445

SciQ

	gpt-oss-20b	gpt-oss-120b	Ministral-8B	Mistral-Nemo-7B	Mistral-Small-3.2-24B	Llama-3.3-70B	Llama-4-Scout-17B	Qwen3-30B-A3B-Instruct	Qwen3-30B-A3B-Thinking	DeepSeek-Llama-70B	DeepSeek-Qwen-32B	gemma-3-27b
ECE	0.9318	0.9227	0.8476	0.8718	0.9214	0.9429	0.9414	0.9341	0.2168	0.9119	0.9068	0.9161
ECE CI	[0.9282–0.9356]	[0.9190–0.9266]	[0.8413–0.8541]	[0.8657–0.8777]	[0.9173–0.9256]	[0.9401–0.9460]	[0.9383–0.9445]	[0.9307–0.9375]	[0.2075–0.2249]	[0.9082–0.9156]	[0.9026–0.9112]	[0.9120–0.9202]
AUROC	0.6237	0.5277	0.5848	0.6313	0.6584	0.5779	0.6017	0.7498	0.6150	0.5003	0.5810	0.6968
Invalid Answers	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000
Invalid UQ Method Scores	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000
Norm. Entropy of Bucket Counts	0.0038	0.1036	0.0000	0.0000	0.0000	0.0000	0.0113	0.0000	0.0000	0.2246	0.1414	0.0159
Accuracy	0.9658	0.9756	0.8809	0.9051	0.9547	0.9762	0.9752	0.9674	0.2501	0.9686	0.9520	0.9501

GPQA

	gpt-oss-20b	gpt-oss-120b	Ministral-8B	Mistral-Nemo-7B	Mistral-Small-3.2-24B	Llama-3.3-70B	Llama-4-Scout-17B	Qwen3-30B-A3B-Instruct	Qwen3-30B-A3B-Thinking	DeepSeek-Llama-70B	DeepSeek-Qwen-32B	gemma-3-27b
ECE	0.7548	0.7638	0.4790	0.5316	0.6347	0.6666	0.6895	0.7398	0.3349	0.6141	0.5588	0.5969
ECE CI	[0.7466–0.7632]	[0.7564–0.7717]	[0.4697–0.4888]	[0.5221–0.5405]	[0.6259–0.6438]	[0.6581–0.6752]	[0.6804–0.6975]	[0.7312–0.7483]	[0.3250–0.3449]	[0.6012–0.6272]	[0.5488–0.5692]	[0.5874–0.6058]
AUROC	0.5094	0.4122	0.4837	0.4963	0.5495	0.6014	0.5661	0.6694	0.4988	0.5366	0.5291	0.5317
Invalid Answers	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/5000	0/10000	0/10000
Invalid UQ Method Scores	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/5000	0/10000	0/10000
Norm. Entropy of Bucket Counts	0.0858	0.1497	0.0000	0.0000	0.0000	0.0000	0.0153	0.0000	0.1876	0.0995	0.1420	0.0007
Accuracy	0.7995	0.8193	0.5123	0.5649	0.6680	0.6999	0.7234	0.7731	0.2513	0.6597	0.6010	0.6303

GSM8K

	gpt-oss-20b	gpt-oss-120b	Ministral-8B	Mistral-Nemo-7B	Mistral-Small-3.2-24B	Llama-3.3-70B	Llama-4-Scout-17B	Qwen3-30B-A3B-Instruct	Qwen3-30B-A3B-Thinking	DeepSeek-Llama-70B	DeepSeek-Qwen-32B	gemma-3-27b
ECE	0.9117	0.9037	0.8610	0.8195	0.9108	0.9362	0.8844	0.9146	0.9137	0.9132	0.9035	0.8743
ECE CI	[0.9019–0.9196]	[0.8944–0.9123]	[0.8484–0.8737]	[0.8046–0.8332]	[0.9021–0.9187]	[0.9292–0.9420]	[0.8749–0.8919]	[0.9065–0.9223]	[0.9054–0.9219]	[0.9046–0.9220]	[0.8938–0.9135]	[0.8641–0.8839]
AUROC	0.6352	0.6483	0.7330	0.7162	0.5887	0.5534	0.5204	0.6349	0.6161	0.6163	0.6036	0.6787
Invalid Answers	4/2500	2/2500	8/2500	12/2500	0/2500	0/2500	0/2500	0/2500	134/2500	0/2500	12/2500	0/2500
Invalid UQ Method Scores	0/2496	0/2498	0/2492	0/2488	0/2500	0/2500	0/2500	0/2500	0/2366	0/2500	0/2488	0/2500
Norm. Entropy of Bucket Counts	0.0160	0.1381	0.0455	0.0367	0.1197	0.0595	0.3760	0.1344	0.1607	0.0382	0.0128	0.2869
Accuracy	0.9459	0.9484	0.8973	0.8549	0.9544	0.9728	0.9660	0.9584	0.9611	0.9492	0.9373	0.9424

SVamp

	gpt-oss-20b	gpt-oss-120b	Ministral-8B	Mistral-Nemo-7B	Mistral-Small-3.2-24B	Llama-3.3-70B	Llama-4-Scout-17B	Qwen3-30B-A3B-Instruct	Qwen3-30B-A3B-Thinking	DeepSeek-Llama-70B	DeepSeek-Qwen-32B	gemma-3-27b
ECE	0.9157	0.8795	0.8393	0.8511	0.8762	0.8999	0.8339	0.8930	0.8759	0.8957	0.9149	0.8700
ECE CI	[0.9074–0.9247]	[0.8706–0.8882]	[0.8278–0.8501]	[0.8387–0.8631]	[0.8663–0.8860]	[0.8909–0.9079]	[0.8230–0.8442]	[0.8847–0.9009]	[0.8662–0.8842]	[0.8862–0.9055]	[0.9066–0.9233]	[0.8618–0.8807]
AUROC	0.6147	0.6179	0.8370	0.7968	0.7863	0.7862	0.7747	0.8250	0.7951	0.6392	0.7172	0.7178
Invalid Answers	4/2500	0/2500	54/2500	39/2500	9/2500	8/2500	1/2500	14/2500	55/2500	201/2500	24/2500	1/2500
Invalid UQ Method Scores	0/2496	0/2500	0/2446	0/2461	0/2491	0/2492	0/2499	0/2486	0/2445	0/2299	0/2476	0/2499
Norm. Entropy of Bucket Counts	0.0999	0.3639	0.4022	0.2187	0.2518	0.2180	0.4590	0.3158	0.3516	0.0392	0.0143	0.3951
Accuracy	0.9559	0.9572	0.9309	0.9065	0.9398	0.9559	0.9400	0.9646	0.9644	0.9317	0.9487	0.9560

SciBench

	gpt-oss-20b	gpt-oss-120b	Ministral-8B	Mistral-Nemo-7B	Mistral-Small-3.2-24B	Llama-3.3-70B	Llama-4-Scout-17B	Qwen3-30B-A3B-Instruct	Qwen3-30B-A3B-Thinking	DeepSeek-Llama-70B	DeepSeek-Qwen-32B	gemma-3-27b
ECE	0.3712	0.3992	0.1487	0.1466	0.4862	0.4272	0.5541	0.6118	0.5579	0.4326	0.4646	0.5150
ECE CI	[0.3522–0.3911]	[0.3802–0.4188]	[0.1331–0.1642]	[0.1310–0.1622]	[0.4661–0.5074]	[0.4068–0.4469]	[0.5357–0.5726]	[0.5928–0.6313]	[0.5361–0.5818]	[0.4114–0.4531]	[0.4425–0.4867]	[0.4949–0.5354]
AUROC	0.5797	0.5451	0.7839	0.7152	0.6379	0.6617	0.6214	0.6631	0.5976	0.5102	0.4619	0.6940
Invalid Answers	45/2500	9/2500	55/2500	60/2500	11/2500	14/2500	9/2500	12/2500	743/2500	328/2500	291/2500	6/2500
Invalid UQ Method Scores	0/2455	0/2491	0/2445	0/2440	0/2489	0/2486	0/2491	0/2488	0/1757	0/2172	0/2209	0/2494
Norm. Entropy of Bucket Counts	0.5178	0.3098	0.0000	0.0000	0.0000	0.0161	0.1976	0.0081	0.0117	0.1033	0.1358	0.0057
Accuracy	0.4134	0.4255	0.1820	0.1799	0.5195	0.4610	0.6078	0.6455	0.5913	0.4337	0.4301	0.5485